When Does Market Access Become Local Development?

Platform Integration, Comparative-Advantage Realization, and Local Value Capture

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Abstract

Platform integration can lower consumer prices everywhere without generating local production and income everywhere. This paper develops a tractable theory of that divergence. A platform simultaneously reallocates household spending from a locally embedded channel toward an external channel and expands the national market accessible to local producers. New Structural Economics disciplines the supply response: regional endowments determine latent comparative advantage; shared infrastructure determines whether an industry consistent with that advantage can cover all economic costs and enter; and a fixed-cost–variable-cost choice determines whether production relies on an external platform organization or becomes embedded in local complementary services. Payment flows, rather than an assumed local-capability index, generate both consumer-channel retention and producer-side value capture. The model yields four results. Stronger latent comparative advantage lowers the infrastructure thresholds for entry and local organizational embeddedness. Nominal and consumption-equivalent local income have distinct infrastructure thresholds; the latter is weakly lower because consumers also benefit from the price decline. Government participation that relaxes a shared-infrastructure financing constraint raises infrastructure supply and reduces the participation needed to cross those thresholds. A separate external-channel restriction can also induce localization, but only by worsening the outside option, thereby reducing consumer access and possibly deterring marginal producers. Illustrative numerical exercises establish that the relevant parameter regions are nonempty; they are not a calibration.

Keywords: platform integration; local value capture; latent comparative advantage; organizational embeddedness; facilitating state; regional development.
JEL codes: F12, O14, O18, R11, L81.

1 Introduction

Digital platforms and domestic market integration can make a national market look more uniform from the consumer's side. A household in a remote place can buy varieties previously available only in a large city, and the effective price of those varieties falls as search, payment, and delivery improve. Evidence from online trade in China shows that e-commerce reduces the role of distance and disproportionately improves consumption access in smaller and more remote cities (Fan et al., 2018). Rural e-commerce expansion also reduced the cost of living for adopters in the experiment studied by Couture et al. (2021). Similar consumer gains arise when modern retail channels enter developing markets (Atkin et al., 2018).

Consumer access, however, is not the same object as local development. The rural e-commerce program studied by Couture et al. (2021) produced little average evidence of higher income for local producers and workers. Modern retail entry can lower the cost of living while displacing income and profits in incumbent local stores (Atkin et al., 2018). More generally, reductions in domestic trade costs do not guarantee that peripheral places industrialize: highway connections to larger markets reduced industrial growth in the peripheral Chinese counties examined by Faber (2014). These findings are not evidence that integration is generally harmful. They show that an improvement in the set of goods a place can buy need not coincide with an improvement in what that place can competitively produce, organize, and retain.

This paper asks when platform-enabled market access becomes local development. Its central distinction is between *access* and *capture*. Access describes the effective prices and market opportunities faced by local consumers and producers. Capture describes the location of the production, service, and profit income generated by the associated transactions. A platform can improve inbound consumer access while shifting local expenditure away from locally embedded wholesale, retail, logistics, and after-sales services. The same platform can improve outbound producer access, but a local industry must still be sufficiently competitive to enter, and it may continue to purchase marketing, fulfillment, certification, settlement, and supply chain services from outside the region after entry. Market participation therefore admits an economically important intermediate state: local production has been realized, but its organization remains externally dependent.

New Structural Economics (NSE) supplies the production-side discipline needed to explain this intermediate state. NSE starts from the proposition that an economy's industrial structure is conditioned by its endowment structure and that each industrial structure requires corresponding hard and soft infrastructure (Lin, 2011, 2012). A firm in an industry consistent with the comparative advantage implied by local factor endowments can nevertheless fail to operate if transaction and coordination costs remain too high. Lin and Wang (2023) distinguish latent comparative advantage (LCA), which is the specialization potential indicated by relative production costs, from actual comparative advantage (ACA), which appears in realized production and exports only after transaction costs are taken into account. This paper uses the domestic-market analogue of that distinction. It does not introduce a free-standing ACA index or treat "realization" as a third comparative- advantage concept. Instead, it asks the narrower equilibrium question required by the platform problem:

how much shared infrastructure is needed for an LCA-consistent industry to enter the national market and then organize local complementary services? The policy distinction matters because shared infrastructure constraints often cannot be removed by a single firm. A facilitating state can help industries with latent comparative advantage overcome infrastructure and coordination constraints (Lin and Monga, 2010); it is not meant to sustain indefinitely an industry that remains nonviable in open competition (Lin, 2003).

The use of NSE here is deliberately narrower than a general theory of structural transformation and broader than relabeling an exogenous "capacity" parameter. Endowments determine relative production costs. Infrastructure affects actual delivered costs and producer market access. Those objects jointly determine whether the LCA-consistent industry is actually produced and sold—the realization margin that Lin and Wang (2023) associate with the movement from LCA to ACA. The organizational form used after entry then determines the location of complementary-service payments. Thus NSE affects both whether local production exists and how much of the resulting value is locally embedded. It does not replace consumer demand, the platform shock, or the local-income propagation mechanism.

The model has five parts. First, a representative household consumes a market good through a local channel and an external platform channel. Platform integration lowers the platform price, raises its expenditure share, and lowers the market-good price index. The local income content of each channel comes from a payment-flow table. When the local channel has greater local income content, the channel shift creates a consumer-side capture drag.

Second, a focal region is embedded in an otherwise exogenous national market. A competitive background industry determines local factor opportunity costs. The region's capital–labor endowment and the candidate industry's factor intensity generate its unit production cost. LCA is defined strictly as a region–industry double relative production cost. Transaction costs, infrastructure, and market size then determine whether that latent cost advantage is realized in equilibrium entry, production, and sales. The nonnegative-profit condition is consistent with Lin (2003)'s viability criterion because fixed costs include the owners' normal return and the model does not use continuing selective protection. Viability is not introduced as another state variable or another threshold.

Third, a representative candidate producer faces CES national demand. The same platform-integration index that reduces the consumer platform price also expands producer market access. Shared infrastructure reduces outbound iceberg costs. A producer can rely on an external platform organization with a low fixed cost and a high unit service payment, or build a locally embedded organization with a higher fixed cost and a lower unit organizational cost. This conditional fixed-cost–variable-cost ordering is related to the trade intermediation mechanism in Ahn et al. (2011). The interpretation of embeddedness as locally supplied specialized producer services follows the development-linkage logic of Rodríguez-Clare (1996). The present paper uses only these two specific ingredients, not either paper's full model.

Fourth, the producer's revenue is split according to the location of actual payments. In the external mode, the unit platform-service payment leaves the region. In the local mode, production and complementary organizational services are locally supplied. These payment flows generate the producer-retention rate; no separate retention-capacity function is assumed. The resulting first-round producer income enters an EL-compatible local-income propagation equation. The propagation block is a one-region, household-endogenous multiplier, closely related to standard social-accounting and local-multiplier reasoning (Miller and Blair, 2009; Moretti, 2010). Its role is conditional: it transmits first-round income generated by the new model rather than constituting the new contribution.

Finally, a reduced-form infrastructure provider faces a financing wedge that falls with government participation. This is a narrow adaptation of the financing mechanism in Lin and Wang (2023), nested within the broader NSE facilitating-state logic in Lin (2011, 2012); Lin and Monga (2010): the state helps relax a financing constraint on shared, transaction-cost-reducing infrastructure. It is not a national social planner, and the paper does not claim that the chosen government participation is nationally optimal.

Four propositions organize the results. Proposition 1 shows that stronger LCA lowers the infrastructure required for profitable entry and actual production. Proposition 2 shows that a second threshold determines organizational embeddedness; stronger LCA also lowers this threshold, and crossing it changes local service income, external payments, and the producer-retention rate. Proposition 3 combines these supply responses with the consumer channel shift. It establishes unique minimum infrastructure thresholds for nonnegative nominal and consumption-equivalent income effects under endpoint-crossing conditions. The real-income threshold is weakly below the nominal-income threshold and is strictly below it when the two roots are interior to the same continuous organizational regime. This qualification matters because discrete entry or organization switches can make the thresholds coincide. Proposition 4 maps the thresholds into minimum government-participation levels. A separate policy example contrasts cost-reducing facilitation with an external-channel restriction.

The paper contributes to three literatures. Relative to the literature on e-commerce and retail access, it explains why consumer price gains can be more geographically diffuse than producer and income gains. Relative to NSE, it introduces platform-mediated organizational dependence as a distinct stage between realizing production and embedding complementary services locally. Relative to local-multiplier models, it derives the first-round local-income term and retention rates from comparative advantage, entry, organization, and payment location. The contribution is not the multiplier itself and not a claim that local retention should be maximized at the expense of consumers.

A companion Economics Letters project in this research program takes first-round producer income and channel retention as given and studies their short-run propagation. The present paper neither repeats that contribution nor uses NSE merely to rename its exogenous local-capability parameter. Instead, the new results are the endowment-based entry threshold, the organizational-embeddedness threshold, and the payment flows that generate first-round local producer income. The multiplier is retained only to make the two papers analytically compatible.

The analysis is intentionally a small-region partial equilibrium. It does not contain migration, land, endogenous national expenditure, platform strategic pricing, or New Economic Geography agglomeration. It therefore cannot evaluate national welfare or long-run spatial equilibrium. This restriction keeps the paper focused on the question that motivates it: when a common platform improves access, what structural conditions allow a region to convert that opportunity into locally captured income?

2 The Access–Capture Benchmark

2.1 Households and consumer access

A representative household allocates nominal income among a market-good composite Q_M, a locally produced nontraded good N, and an outside numeraire Z:

$$U = Q_M^{\alpha_M} N^{\beta} Z^{1-\alpha_M-\beta}, \qquad \alpha_M > 0, \quad \beta \geq 0, \quad \alpha_M + \beta < 1. \tag{1}$$

Following the standard CES benchmark (Dixit and Stiglitz, 1977), the market-good composite is purchased through a locally embedded channel A and an external platform channel P:

$$Q_M = \left[(1-a_P)^{1/\eta} Q_A^{(\eta-1)/\eta} + a_P^{1/\eta} Q_P^{(\eta-1)/\eta}\right]^{\eta/(\eta-1)}, \qquad \eta > 1. \tag{2}$$

Normalize $p_A = 1$. Platform integration z lowers the effective platform price:

$$p_P(z) = \bar{p}_P e^{-z}. \tag{3}$$

This price can include the delivered product price, search and payment costs, and any consumer-facing platform charge. It is distinct from the producer-side organizational payment introduced below.

The platform expenditure share and market-good price index are:

$$s(z) = \frac{a_P p_P(z)^{1-\eta}}{(1-a_P)p_A^{1-\eta} + a_P p_P(z)^{1-\eta}}, \tag{4}$$

$$P_M(z) = \left[(1-a_P)p_A^{1-\eta} + a_P p_P(z)^{1-\eta}\right]^{1/(1-\eta)}. \tag{5}$$

Standard CES differentiation gives:

$$s'(z) = (\eta-1)s(z)[1-s(z)] > 0, \qquad \frac{d\ln P_M(z)}{dz} = -s(z) < 0. \tag{6}$$

Thus the platform unambiguously improves the model's consumer-access object: its expenditure share rises and the cost of a unit of Q_M falls.

2.2 Consumer-channel payment incidence

Access does not determine where income is generated. Let ℓ_A and ℓ_P denote the local income generated per unit of expenditure through the two consumer channels. Table 1 clarifies their content. Both measures use the same transaction boundary and therefore can be combined without treating an external payment as a national resource loss. Formally, for $r \in \{A, P\}$:

$$\ell_r = \frac{LI_r}{E_r}, \qquad 0 \leq \ell_r \leq 1. \tag{7}$$

Here E_r is gross local-household expenditure through channel r, and LI_r is the associated local factor income plus locally owned operating surplus plus local service value added. Taxes and transfers are excluded because the model contains no government budget or interregional transfer rule.

Table 1: Consumer-channel payment incidence

Payment generated by one unit of local household expenditure	Local channel A	External platform P
Local wholesale, retail, storage, and after-sales value added	local income	limited in baseline
External production and interregional shipping	external payment	external payment
Platform, settlement, and externally supplied digital services	limited	external payment
Locally owned operating surplus	local income	included if locally owned
Local income content	ℓ_A	ℓ_P

The baseline assumes $\ell_A > \ell_P$. This is a transparent spatial incidence assumption, not an assertion that platforms always retain less value locally. A platform that develops local fulfillment, seller services, or ownership can have $\ell_P \geq \ell_A$; that reverse case is reported in the appendix.

The local income content of household market-good expenditure is:

$$\omega_C(z) = [1 - s(z)]\ell_A + s(z)\ell_P. \tag{8}$$

Define:

$$D(z) = 1 - \beta - \alpha_M \omega_C(z). \tag{9}$$

Because $\omega_C(z) \leq 1$ and $\alpha_M + \beta < 1$, $D(z) \geq 1 - \alpha_M - \beta > 0$; no additional stability assumption is required. When $\ell_A > \ell_P$, platform substitution lowers ω_C and raises D. The corresponding proportional capture drag is:

$$\Lambda(z) = \frac{\alpha_M(\ell_A - \ell_P)s'(z)}{D(z)} > 0. \tag{10}$$

The numerator records how rapidly expenditure moves toward the lower-retention channel; the denominator captures repeated local spending. This term is consumer-side channel displacement. It is separate from the producer-side external platform payment developed in Section 4.

3 Endowments, Comparative Advantage, and Producer Access

3.1 A focal region and factor opportunity costs

Consider a small focal region i with baseline factor endowments K_i and L_i. Let:

$$e_i = \frac{K_i}{L_i}. \tag{11}$$

A competitive background industry 0 uses:

$$Y_{i0} = A_0 K_{i0}^{\theta_0} L_{i0}^{1-\theta_0}. \tag{12}$$

The candidate industry is small relative to the background economy. The background industry therefore determines the factor opportunity costs relevant for its entry decision. Over the range studied, marginal factor services can be supplied at those opportunity costs without changing the factor-price schedule. This assumption can represent underused capacity, additional work hours, or a locally elastic short-run supply margin. With the background good as numeraire, competitive factor pricing gives:

$$w_i = (1-\theta_0)A_0 e_i^{\theta_0}, \qquad r_i = \theta_0 A_0 e_i^{\theta_0-1}. \tag{13}$$

This closure avoids treating factor prices as free parameters while preserving a tractable small-region problem. It is not a full-employment two-sector general equilibrium: a long-run welfare analysis would have to subtract production forgone in the background industry and solve the factor reallocation explicitly.

Candidate industry j uses:

$$Y_{ij} = A_j K_{ij}^{\theta_j} L_{ij}^{1-\theta_j}. \tag{14}$$

Its unit production cost is:

$$\begin{aligned} c_{ij}^P &= \frac{1}{A_j}\left(\frac{r_i}{\theta_j}\right)^{\theta_j}\left(\frac{w_i}{1-\theta_j}\right)^{1-\theta_j} \\ &= \xi_j e_i^{\theta_0-\theta_j}, \qquad \xi_j > 0. \end{aligned} \tag{15}$$

If the candidate industry is more capital intensive than the background industry, $\theta_j > \theta_0$, a higher regional capital–labor endowment lowers its relative unit cost. This is the minimal factor-proportions channel required for an NSE paper. It is consistent with the broader result that factor abundance shifts production and trade toward industries intensive in the abundant factor (Romalis, 2004), and with the NSE view that industrial composition evolves with endowment structure (Ju et al., 2015).

3.2 Latent comparative advantage and actual realization

The NSE sources do not posit a four-stage sequence consisting of separate LCA, ACA, viability, and realization variables. The distinction used by Lin and Wang (2023) has two economic margins. Relative production costs, predicted by endowments and technology, identify an industry's LCA. Actual production and exports depend on total costs, including the fixed and variable transaction costs affected by hard and soft infrastructure; the realized production and export pattern is ACA. In their empirical analysis, LCA is proxied by the distance between national factor endowments and product factor requirements, while ACA is observed through export participation and revealed comparative advantage. In their model, infrastructure changes transaction costs and hence the set of active industries. ACA and "realization" are therefore not two additional concepts to be modelled separately.

Definition 1 (Latent comparative advantage)**.** In the terminology of Lin and Wang (2023), an industry has latent comparative advantage when the location has a relative production-cost advantage rooted in its endowment structure, before the transaction costs that determine actual competitiveness are added. This definition combines the NSE proposition that endowment structure determines comparative advantage (Lin, 2011, 2012; Ju et al., 2015) with the production-cost/transaction-cost distinction in Lin and Wang (2023).

Let h be an external reference region and 0 a background industry in each region. The present paper represents the relative-production-cost criterion by:

$$\mathcal{C}_{ij}^{P} = \frac{c_{ij}^{P}/c_{i0}^{P}}{c_{hj}^{P}/c_{h0}^{P}}. \tag{16}$$

Region i has LCA in industry j when $\mathcal{C}_{ij}^{P} < 1$.

Equation (16) is this paper's two-region, two-industry implementation of the comparative-cost criterion; it is not a formula quoted from Lin and Wang (2023). The double comparison is necessary because a low absolute cost or a positive accounting margin is not comparative advantage. The factor-proportions mechanism behind the relative cost is consistent with Romalis (2004), while the interpretation of that cost pattern as endowment-conditioned industrial potential follows NSE. In the analytical and numerical comparative statics below, $c \equiv c_{ij}^{P}$ is only a monotone sufficient statistic for $\mathcal{C}_{ij}^{P}$, holding the reference-region and background-industry ratios fixed:

$$\frac{\partial c}{\partial \mathcal{C}_{ij}^{P}} > 0. \tag{17}$$

Lower c therefore means stronger LCA within the stated comparison, but the underlying definition remains equation (16).

The model does not construct a second, free-standing ACA index. Such an index would require total costs and organizational alternatives for both the focal and reference regions, objects that are unnecessary in a focal-region model. Instead, infrastructure G changes the focal region's transaction costs and producer access, and the producer problem determines whether the candidate industry is actually active. When

$$\max\{\pi_P, \pi_L\} \geq 0,$$

the LCA-consistent industry enters, produces, and sells through an available organization. This event is the model's domestic-market counterpart of the LCA-to-ACA realization in Lin and Wang (2023); it is not a new definition of comparative advantage.

The related concept of viability comes from a different NSE argument. Lin (2003) defines a normally managed firm as viable when it can earn a socially acceptable normal profit in a free, open, and competitive market without external subsidy or protection. The nonnegative-profit entry condition above is compatible with that criterion once fixed costs include the owners' normal return and shared infrastructure is distinguished from open-ended firm-specific support. It is not a complete separate model of viability, however, and the paper introduces neither a viability index nor a viability threshold distinct from entry.

The theoretical mapping used below is therefore:

$$\underbrace{(E_i, \theta_j) \longmapsto c_{ij}^P / c_{i0}^P}_{\text{LCA}} \quad + \quad \underbrace{(G, \tau_O, \mathcal{M})}_{\text{realization constraints}} \quad \Longrightarrow \quad \underbrace{\max_{r \in \{P,L\}} \pi_r \geq 0}_{\text{actual production / ACA}} \quad .$$

Observed entry alone cannot identify the counterfactual production-cost comparison required for LCA, but the model needs no intervening ACA or viability state. The formal threshold G_E below is simply the minimum infrastructure level for actual entry.

3.3 Producer market access

The candidate producer faces a national CES demand schedule:

$$q_r = \mathcal{M}(z, G) p_r^{-\sigma}, \qquad \sigma > 1. \tag{18}$$

Outbound iceberg costs are:

$$\tau_O(z, G) = \bar{\tau}_O \exp[-\psi z - \gamma G], \qquad \bar{\tau}_O \geq 1, \quad \psi > 0, \quad \gamma > 0. \tag{19}$$

Here z lowers platform-mediated search, coordination, and delivery costs, while G is common, non-firm-specific hard or soft infrastructure such as a transport link, broadband backbone, public standard, certification system, or contract-enforcement facility. The local analysis is restricted to the range in which the iceberg representation remains at least one.

Substituting equation (19) into CES demand and absorbing the invariant factor $\bar{\tau}_O^{1-\sigma}$ and national expenditure into $\mathcal{M}_0$ gives:

$$\mathcal{M}(z, G) = \mathcal{M}_0 \exp\left\{(\sigma - 1)(\psi z + \gamma G)\right\}. \tag{20}$$

This is the standard market-access mapping in which iceberg trade costs enter CES demand with elasticity $1 - \sigma$ (Donaldson and Hornbeck, 2016). The baseline deliberately excludes a direct zG complementarity: both reforms lower the same outbound cost, and no additional parameter is needed for the core results. The appendix considers the extension $\tau_O = \bar{\tau}_O \exp[-\psi z - \gamma G - \chi^{\text{ext}} zG]$, $\chi^{\text{ext}} > 0$.

Equation (20) is not the consumer price equation written twice. Consumer access in equation (3) concerns what local households can buy; producer access concerns the demand that local producers can reach. The same platform reform z moves both, which keeps the EL question intact, but the two margins have different payment incidence and only producer access depends on G.

3.4 Entry and the actual-entry threshold

The producer can use an external platform organization P or a locally embedded organization L:

$$m_P = c + d_P, \qquad m_L = c + d_L, \qquad 0 \leq d_L < d_P, \tag{21}$$

$$F_P = f, \qquad F_L = f + F, \qquad f > 0, \quad F > 0. \tag{22}$$

The external organization uses a platform's existing national marketing, settlement, fulfillment, and supply-chain system. It therefore avoids the organization-specific fixed cost F, but it entails a larger unit service payment. The common entry cost f is a private economic cost of product adaptation, certification, and initial market entry; it includes the normal return required for a zero-economic-profit entry boundary. The additional F is the private or collectively financed setup cost of a local warehouse, operations team, quality-control system, or seller-service network. It is distinct from G, which is non-firm-specific shared infrastructure. The local organization builds or coordinates the complementary services represented by F locally. This ordering is conditional, not a universal property of digital platforms. It isolates a single organizational choice: buy an external service flow or incur a fixed setup cost to embed that service locally.

With constant elasticity demand, the markup, price, revenue, and profit are:

$$\mu = \frac{\sigma}{\sigma - 1}, \qquad\qquad p_r = \mu m_r, \tag{23}$$

$$\mathcal{R}_r(z, G) = \mathcal{M}(z, G)\mu^{1-\sigma} m_r^{1-\sigma}, \qquad\qquad \pi_r(z, G) = \frac{\mathcal{R}_r(z, G)}{\sigma} - F_r. \tag{24}$$

The producer selects:

$$r^*(z, G) \in \arg\max\{0, \pi_P(z, G), \pi_L(z, G)\}. \tag{25}$$

Define:

$$k = \frac{\mu^{1-\sigma}}{\sigma}, \qquad \Omega = (\sigma - 1)\gamma > 0. \tag{26}$$

The zero-profit infrastructure threshold for organization r is:

$$G_r^0(z,c) = \frac{\ln\left[\dfrac{F_r}{k\mathcal{M}_0 e^{(\sigma-1)\psi z}(c+d_r)^{1-\sigma}}\right]}{\Omega}. \tag{27}$$

The economically relevant entry threshold is:

$$G_E(z,c) = \max\left\{0, \min[G_P^0(z,c), G_L^0(z,c)]\right\}. \tag{28}$$

Proposition 1 (LCA and actual entry)**.** *Suppose $\gamma > 0$. The minimum infrastructure level G_E for profitable entry exists. At an interior threshold:*

$$\frac{\partial G_E}{\partial c} > 0, \qquad \frac{\partial G_E}{\partial \mathcal{C}_{ij}^P} > 0. \tag{29}$$

Hence stronger LCA lowers the shared infrastructure required for profitable entry and actual production. If a raw threshold is negative, its economically truncated value is zero and the comparative static is weak.

The result is not an assumption that LCA raises local income. It follows from the profit problem: a lower endowment-conditioned production cost raises operating profit under either organization and reduces the amount of transaction-cost-reducing infrastructure needed to cover the fixed cost. Proofs of all propositions are in the appendix.

4 Organization, Capture, and Local Income

4.1 The organizational-embeddedness threshold

The local mode has a higher operating-profit coefficient because $d_L < d_P$, but it must cover the additional fixed cost F. The infrastructure level at which $\pi_L = \pi_P$ is:

$$G_X(z,c) = \frac{\ln\left[\dfrac{F}{k\mathcal{M}_0 e^{(\sigma-1)\psi z}\{(c+d_L)^{1-\sigma} - (c+d_P)^{1-\sigma}\}}\right]}{\Omega}. \tag{30}$$

The economically relevant embeddedness threshold is defined directly from the equilibrium choice:

$$G_L(z,c) = \inf\left\{G \geq 0 : \pi_L(z,G) \geq \max\{0, \pi_P(z,G)\}\right\}. \tag{31}$$

The full state classification requires distinguishing G_P^0, G_L^0, and G_X. Let:

$$b_r = k(c+d_r)^{1-\sigma}, \qquad \bar{F} = f\frac{b_L - b_P}{b_P}. \tag{32}$$

If $F > \bar{F}$, then:

$$G_P^0 < G_L^0 < G_X. \tag{33}$$

The equilibrium consequently passes through three regions as infrastructure rises:

$$\text{no entry} \quad \longrightarrow \quad \text{external platform dependence} \quad \longrightarrow \quad \text{local organizational embeddedness.}$$

If $F \leq \bar{F}$, the platform-dependent middle region can disappear: the producer may move directly from no entry to the local organization. The middle state is therefore an economic result with a transparent parameter condition, not a stage imposed by terminology. Equivalently:

$$G_L(z,c) = \begin{cases} \max\{0, G_X(z,c)\}, & F > \bar{F}, \\ \max\{0, G_L^0(z,c)\}, & F \leq \bar{F}. \end{cases} \tag{34}$$

4.2 Producer-side payment incidence

Producer revenue is not automatically local income. Let $d_P q_P$ be the payment to the external platform and externally supplied complementary services. Production factors and the residual operating surplus are locally owned in the baseline; imported intermediate inputs and outside equity are excluded from the polar benchmark. Under the local mode, the organizational-service payment $d_L q_L$ is made to local providers. Table 2 records the incidence.

Table 2: Producer organization and spatial payment incidence

Payment generated by candidate-industry sales	External mode P	Local mode L
Local production-factor income	local	local
Local producer operating surplus	local	local
Unit marketing, settlement, fulfillment, and supply-chain service	external $d_P q_P$	local $d_L q_L$
Shared organizational fixed cost	not incurred	locally supplied F
Common entry cost	locally supplied f	locally supplied f
External payment included in retention measure	$d_P q_P$	0

Fixed costs affect entry and organization, but locally supplied fixed-cost services are payments to local factors or firms. Deducting them again from local income would double count a local resource payment. Conversely, the external unit payment is excluded from local income even if it is not a national loss. Equivalently, first-round local income is local factor income plus locally owned operating surplus plus locally supplied producer-service value added, all measured within the same transaction boundary.

Let B^r_j denote first-round local income generated by candidate-industry sales:

$$B^r_j(z, G) = \rho_r \mathcal{R}_r(z, G). \tag{35}$$

The payment table implies:

$$\rho_P = 1 - \frac{d_P}{\mu(c + d_P)} < 1, \qquad \rho_L = 1. \tag{36}$$

The value $\rho_L = 1$ is a polar incidence benchmark, not a claim that every locally organized supply chain has no outside payment. It follows here from local ownership of production and organizational services and the exclusion of imported intermediates. Allowing such payments would lower ρ_L without changing the threshold logic provided the local mode remains more locally embedded than the external mode. If a fraction $\lambda \in [0, 1]$ of platform services is supplied locally:

$$\rho_P(\lambda) = 1 - \frac{(1-\lambda)d_P}{\mu(c + d_P)}. \tag{37}$$

The baseline is $\lambda = 0$. This robustness parameter changes payment location, not productive efficiency.

Proposition 2 (Organizational embeddedness)**.** *Suppose $d_L < d_P$, $F > 0$, and $\Omega > 0$. There exists a minimum embeddedness threshold G_L. At an interior threshold:*

$$\frac{\partial G_L}{\partial c} > 0, \qquad \frac{\partial G_L}{\partial \mathcal{C}^P_{ij}} > 0. \tag{38}$$

For $G \geq G_L$, the producer selects L, and the switch satisfies:

$$\rho_L > \rho_P, \qquad \mathcal{R}_L > \mathcal{R}_P, \qquad B^L_j > B^P_j. \tag{39}$$

If $F > \bar{F}$, platform dependence is a distinct equilibrium region between entry and local embeddedness. If $F \leq \bar{F}$, that region may disappear.

This proposition is the main way NSE affects capture beyond production. Stronger LCA makes local sales sufficiently large to cover the shared organizational fixed cost at a lower infrastructure level. Crossing the threshold does not merely relabel the firm: it reduces external payments and creates local complementary-service income.

4.3 Nominal and consumption-equivalent local income

Let $B_0 > 0$ denote autonomous first-round income from the background economy over the local partial-equilibrium range. Consistent with the locally elastic factor-supply assumption above, candidate-industry factor payments are incremental nominal receipts; their full opportunity cost is not a welfare gain. Household spending generates additional rounds of local income. The share β

spent on locally produced nontraded goods remains local. The share α_M spent on market goods generates local income at rate $\omega_C(z)$. Avoiding any duplication of the first-round producer payments, nominal local income satisfies:

$$Y = B_0 + B_j^{r^*}(z, G) + \beta Y + \alpha_M \omega_C(z) Y. \tag{40}$$

Therefore:

$$Y(z, G) = \frac{B_0 + B_j^{r^*}(z, G)}{D(z)}. \tag{41}$$

This is a conditional local-income accounting equilibrium. It is appropriate when the relevant local services and nontraded goods can respond at a fixed short-run opportunity cost. It is not a full welfare measure.

Consumption-equivalent income deflates nominal income by the market-good price index:

$$R(z, G) = \frac{Y(z, G)}{P_M(z)^{\alpha_M}}. \tag{42}$$

Define the candidate industry's share in first-round local income:

$$\zeta_r(z, G) = \frac{B_j^r(z, G)}{B_0 + B_j^r(z, G)}. \tag{43}$$

Within a fixed organizational state, equations (20), (10), and (41) yield:

$$\frac{d \ln Y}{dz} = \zeta_r(z, G)(\sigma - 1)\psi - \Lambda(z). \tag{44}$$

The first term is the local producer-income gain from better platform access. Its direct elasticity $(\sigma - 1)\psi$ is common across regions, but its local-income effect is larger when infrastructure has already raised the candidate industry's share ζ_r. The second term is the consumer-channel capture drag.

The consumption-equivalent effect is:

$$\frac{d \ln R}{dz} = \frac{d \ln Y}{dz} + \alpha_M s(z). \tag{45}$$

The final term is the direct consumer price benefit. Equations (44) and (45) preserve the original access–capture question while replacing an exogenous producer gain and retention capacity with equilibrium entry, organization, and payment flows.

Proposition 3 (Nominal and real-income thresholds)**.** *Suppose $\psi > 0$ and maintain the baseline payment ordering. Within each active producer organization, both $d \ln Y/dz$ and $d \ln R/dz$ are strictly increasing in G; in the no-entry state they are constant in G. Entry and the switch from P to L generate upward jumps. If the relevant effect is negative at the lower endpoint and positive at* *an upper endpoint, there is a unique minimum threshold:*

$$G_Y = \inf \left\{ G : \frac{d \ln Y}{dz} \geq 0 \right\}, \tag{46}$$

$$G_R = \inf \left\{ G : \frac{d \ln R}{dz} \geq 0 \right\}. \tag{47}$$

Pointwise,

$$G_R \leq G_Y. \tag{48}$$

If both roots are interior to the same continuous organizational region,

$$G_R < G_Y. \tag{49}$$

If both effects cross zero at the same discrete entry or organization switch, the thresholds can coincide. Stronger LCA weakly lowers both thresholds and strictly lowers them when the relevant roots remain interior to the same organizational states.

In the unbounded-G baseline, sufficient upper-endpoint conditions are $(\sigma - 1)\psi > \Lambda(z)$ for the nominal threshold and $(\sigma - 1)\psi + \alpha_M s(z) > \Lambda(z)$ for the real threshold. These conditions are substantive: after removing the unnecessary zG interaction, the existence of a favorable high-infrastructure endpoint is no longer built into the functional form.

The proposition separates three regional outcomes. Below G_R, producer gains are too small to offset both channel displacement and the relevant cost of living, so nominal and consumption-equivalent income fall. Between G_R and G_Y, nominal income falls but the consumer price gain is large enough to raise consumption-equivalent income. Above G_Y, producer access and local capture are strong enough for both measures to rise. The classification is conditional: if $\ell_P \geq \ell_A$, capture drag is nonpositive and the low-infrastructure loss region can disappear.

5 The Facilitating State

5.1 Infrastructure supply

The preceding analysis treats G as a local structural condition. This section endogenizes its supply without introducing a national social planner. Let $\vartheta \in [0, 1]$ denote government participation in infrastructure finance. The financing wedge is:

$$\kappa_I(\vartheta) = \kappa_p - (\kappa_p - \kappa_g)\vartheta, \qquad 0 < \kappa_g < \kappa_p. \tag{50}$$

The infrastructure provider solves:

$$\max_{G \geq 0} \left\{ \mathcal{U}_I(G) - \kappa_I(\vartheta) C(G) \right\}, \tag{51}$$

where:

$$\mathcal{U}_I' > 0, \quad \mathcal{U}_I'' < 0, \quad C' > 0, \quad C'' \geq 0. \tag{52}$$

$\mathcal{U}_I$ is the provider's appropriable revenue from infrastructure services. The gap between this revenue and the broader producer-access gains captures the shared nature of logistics, standards, certification, digital systems, and other infrastructure. Government participation lowers the financing wedge; it does not directly subsidize candidate-industry output.

An interior solution satisfies:

$$\mathcal{U}_I'[G(\vartheta)] = \kappa_I(\vartheta)C'[G(\vartheta)]. \tag{53}$$

The curvature assumptions guarantee uniqueness.

5.2 Government-participation thresholds

Proposition 4 (Government participation and threshold crossing)**.** *The infrastructure supply in equation* (51) *is unique and:*

$$G'(\vartheta) > 0. \tag{54}$$

For $H \in \{E, L, R, Y\}$, define the minimum government participation needed to cross an existing threshold:

$$\vartheta_H = \inf\{\vartheta : G(\vartheta) \geq G_H\}. \tag{55}$$

Greater government participation reduces each gap $G_H - G(\vartheta)$. Stronger LCA weakly lowers $\vartheta_E, \vartheta_L, \vartheta_R$, and ϑ_Y, with strict effects where the corresponding structural thresholds are interior.

5.3 Cost-reducing facilitation versus an external-channel restriction

The preceding proposition concerns only the supply of shared infrastructure. It should not be read as a theorem that every state intervention is facilitating. To make the distinction concrete, consider a narrow external-channel restriction that multiplies p_P by a wedge $\tau_C > 1$ and raises d_P. This is not a general definition of protection in NSE; it is a policy example tailored to the two channels in this model. The consumer price index rises and the platform expenditure share falls. On the producer side:

$$\frac{\partial G_P^0}{\partial d_P} = \frac{\sigma - 1}{\Omega(c + d_P)} > 0. \tag{56}$$

The restriction therefore makes external-platform entry harder. At the same time, raising d_P increases the local mode's variable-cost advantage and can lower G_X. It can induce localization by degrading the alternative. Facilitation instead allows the region to cross the entry and embeddedness thresholds by reducing a real shared constraint. This conditional comparison is consistent with the NSE distinction between facilitating an LCA-consistent, potentially viable industry and protecting a nonviable activity; it is not a complete ranking of all possible protection and facilitation policies.

The state result is deliberately limited. It does not prove that more government participation is always desirable, that the provider captures all social returns, or that $\vartheta = 1$ is optimal. A complete national welfare problem would require government fiscal costs, external platform profits, other regions, and general-equilibrium factor responses, all of which are outside the focal-region model.

6 Illustrative Numerical Exercises

6.1 Purpose and parameterization

The numerical exercises have one purpose: to demonstrate that the parameter regions described by Propositions 1–4 are nonempty and to display the model's discontinuities. They are not a calibration and none of the values is presented as an estimate for China. Table 3 reports the standardized baseline.

Table 3: Illustrative baseline parameters

Block	Parameters	Values
Household	α_M, β, η	0.35, 0.30, 6
Consumer channels	ℓ_A, ℓ_P, B_0	0.70, 0.10, 1
Producer demand	$\sigma, \mathcal{M}_0, \psi$	4, 5, 0.19
Infrastructure access	γ	0.10
Production and organization	c, d_P, d_L, f, F	0.9, 0.4, 0.1, 0.34, 0.53
State block	κ_p, κ_g, a_I	4.0, 0.30, 0.80

The consumer CES weight is $a_P = 0.4$, and the pre-integration platform price is $\bar{p}_P = 1.2$. For the state exercise:

$$\mathcal{U}_I(G) = a_I \ln(1+G), \qquad C(G) = \frac{G^2}{2}. \tag{57}$$

Equation (53) then has the closed-form solution:

$$G(\vartheta) = \frac{-1 + \sqrt{1 + 4a_I/\kappa_I(\vartheta)}}{2}. \tag{58}$$

6.2 The structural phase diagram

Figure 1 varies the candidate industry's relative production cost and infrastructure at $z = 0.5$. Moving right weakens LCA. Moving up improves the infrastructure through which LCA can be converted into actual entry and local organization. The colored regions are filled directly between the analytical boundaries $G_E(c)$ and $G_L(c)$; they are not a smoothed or interpolated classification grid.

The gray region contains no active candidate industry. The blue region contains local production that reaches the national market through the external platform organization. The orange region contains locally embedded production and complementary services. At a given G, a stronger LCA industry is more likely both to enter and to embed. For an intermediate range of production costs,

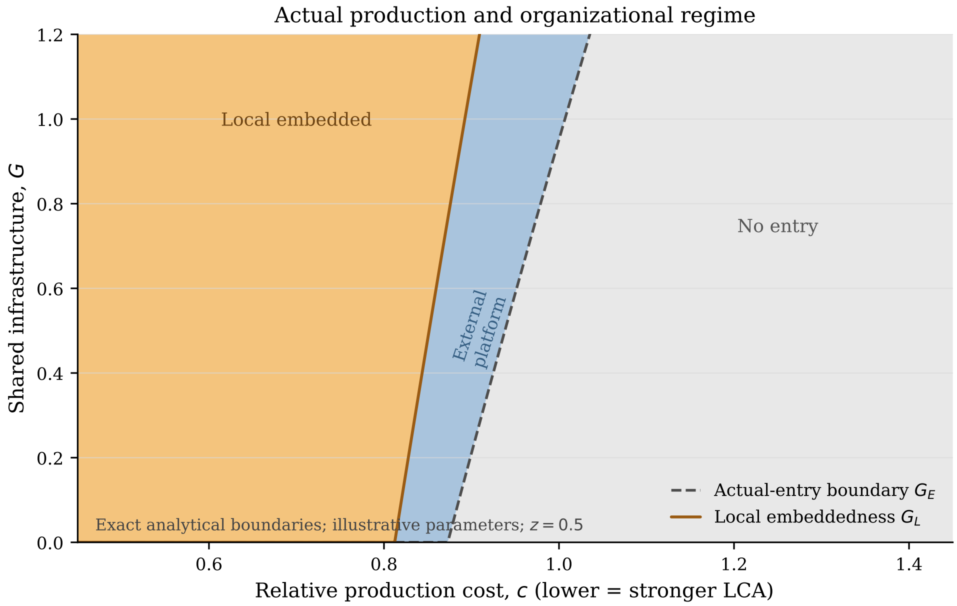

Figure 1: Analytical phase diagram for actual production and organization

increasing infrastructure moves the region through all three states; sufficiently strong LCA can make the industry locally embedded even at low G, whereas sufficiently weak LCA may leave it unprofitable throughout the plotted range. The platform-dependent middle region is sizable under the baseline because $F > \bar{F}$; it is generated by the profit and organization problem rather than imposed as an additional comparative- advantage category.

6.3 Infrastructure thresholds

Figure 2 separates the structural and income thresholds that were previously superimposed in one panel. Panel (a) plots actual entry and local embeddedness; its shaded band is the platform-dependent interval. Panel (b) plots the real- and nominal-income thresholds; its shaded band is the range in which the real effect is nonnegative but the nominal effect remains negative. All curves are computed from the exact equilibrium choice and the minimum-G definitions in equations (28), (31), (46), and (47).

All four thresholds weakly rise as LCA weakens. Flat zero segments and kinks are economically meaningful: they arise from the nonnegativity bound on infrastructure or from a change in the active equilibrium mode. They have not been removed by statistical smoothing. In the baseline at $c = 0.9, z = 0.5$, the relevant thresholds are:

$$G_E = G_R = 0.2106, \qquad G_Y = 0.5554, \qquad G_L = 1.0910.$$

Here platform entry produces a discrete first-round income gain large enough to turn the real effect positive, so the economic infimum satisfies $G_R = G_E$ exactly. The nominal effect turns positive

LCA-conditioned infrastructure thresholds

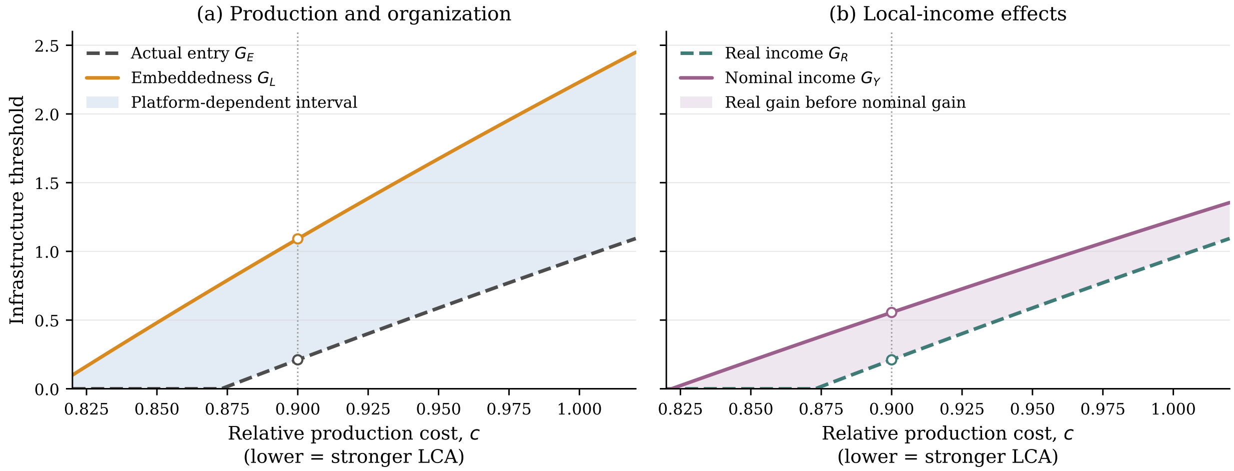

Figure 2: LCA-conditioned infrastructure thresholds

only at an interior root. The strict ordering $G_R < G_Y$ is numerical; the proposition preserves the weak ordering when both income thresholds are governed by the same discrete switch.

6.4 Platform integration under alternative state participation

The state specification yields:

$$G(0.05) = 0.1780, \qquad G(0.50) = 0.2887, \qquad G(0.99) = 1.1198.$$

Figure 3 follows platform integration for those three infrastructure levels. The first panel is common across participation levels because the baseline state intervention affects producer infrastructure, not the consumer platform price. The second and third panels report nominal and consumption-equivalent income relative to their own $z = 0$ values. The curves are drawn separately within each equilibrium regime. At entry and organization switches, open and filled markers report the left- and right-limit values and a vertical segment records the discrete jump. Hence the figure does not connect a structural discontinuity with an artificial diagonal line. The source-data files also report the transition events and the underlying marginal effects used in Proposition 3.

The vertical jumps are economic features of the representative-industry baseline, not insufficient plot resolution. Crossing G_E adds the candidate industry's entire first-round income, while crossing G_L changes the location of its complementary-service payments. Their numerical size depends on the standardized ratio of candidate-industry demand to background income and is not an empirical effect size. A continuum of heterogeneous firms would distribute the entry and organization cutoffs and thereby smooth aggregate income paths, but it would add a productivity distribution and would remove the closed-form representative-industry benchmark. Applying a spline to the present model would instead be incorrect because it would fabricate intermediate equilibria.

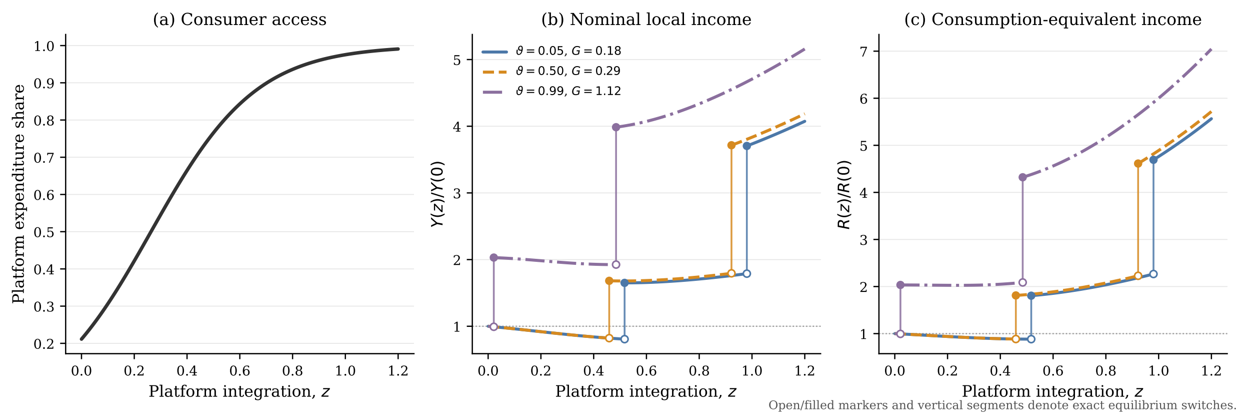

Figure 3: Event-aware platform-integration and local-income paths

Platform penetration rises for every region. At low participation, the candidate industry enters through P around $z = 0.52$, reversing the pre-entry decline in both income paths; it switches to L only near $z = 0.98$. At middle participation, entry occurs earlier, around $z = 0.46$; consumption-equivalent income begins rising at entry, while nominal income reaches its local minimum around $z = 0.52$, and local embeddedness occurs near $z = 0.92$. At high participation, entry occurs almost immediately, around $z = 0.02$, and the producer switches to L around $z = 0.49$. Before that switch, consumer-side capture drag can keep the nominal path gently declining; the discrete increase at the switch is the endogenous local-service and payment-incidence effect. The consumer platform share is common across cases because the baseline state intervention changes producer infrastructure, not the consumer platform price.

The implied government-participation thresholds at $z = 0.5, c = 0.9$ are:

$$\vartheta_E = \vartheta_R = 0.2331, \quad \vartheta_Y = 0.8308, \quad \vartheta_L = 0.9863.$$

Thus the revised parameterization makes every structural threshold interior to the admissible participation interval rather than mechanically assigning most of them a zero state-participation requirement.

The appendix reports six mechanism checks. Setting $\ell_A = \ell_P$ eliminates consumer-side capture drag. Equating ρ_P and ρ_L eliminates the producer-side spatial payment difference. A positive χ^{ext} adds, rather than removes, an infrastructure–platform complementarity to the parsimonious baseline. A positive λ partially localizes platform services. The reverse case $\ell_P > \ell_A$ turns capture drag into a local retention gain. Finally, an equal-localization comparison shows that an external-channel restriction and facilitation have different consumer-access and marginal-entry consequences. Every figure is generated from the same public code and is accompanied by CSV source data.

7 Discussion and Testable Implications

7.1 What the model does and does not call development

The model distinguishes three outcomes. P_M measures consumer access. Y measures nominal local income generated within the focal accounting boundary. R measures how much market-good consumption that nominal income can support after the platform price change. None is a national welfare measure. A payment to an external domestic platform is a leakage from the focal region but remains income somewhere in the national economy. Likewise, a high local-retention rate is not itself a sufficient policy objective: a policy can mechanically raise retention by making an efficient external channel more costly.

This distinction prevents a protectionist reading of the access–capture problem. A region with little infrastructure may receive a genuine consumer gain even when nominal local income falls. The relevant development question is not how to preserve every incumbent local intermediary. It is how to enable industries consistent with the region's latent comparative advantage to enter, expand, and build complementary local services without sacrificing the consumer gains from national integration.

7.2 Testable implications

Although the paper contains no original data, its mechanisms generate specific empirical predictions.

First, platform expansion should reduce consumer price dispersion more broadly than it increases local producer income. Consumer effects should depend primarily on pre-platform access and platform adoption, whereas producer-income effects should load on the interaction between pre-existing endowment–industry congruence and infrastructure. This prediction sharpens the access–income divergence documented by Couture et al. (2021).

Second, among locations with similar platform access, lower endowment-conditioned relative production costs should predict earlier seller entry. Infrastructure should have larger producer effects in industries that are intensive in locally abundant factors. A credible test would construct LCA from pre-shock regional endowments and national industry factor intensities, rather than infer comparative advantage from post-shock sales.

Third, producer entry and local capture should not be treated as the same outcome. In the platform-dependent region, local seller sales can rise while platform, settlement, fulfillment, certification, and marketing payments remain external. Crossing the embeddedness threshold predicts a discrete increase in local complementary-service establishments, employment, or payroll relative to sales. Buyer–seller locations alone are therefore insufficient to measure capture; service-provider and ownership locations are also needed.

Fourth, facilitating infrastructure and an external-channel restriction should have different joint signatures. Facilitation should raise outbound seller participation without raising the local consumer platform price. An external-channel restriction may increase the local share of some services while reducing platform adoption, raising consumer prices, and deterring marginal platform-dependent sellers. Empirically similar local-sales shares can therefore conceal very different access and productivity consequences.

Fifth, partial localization of platform services should compress the difference between P and L. Regions hosting fulfillment, seller operations, payment, or platform-service employment should retain more value for a given volume of platform trade. This prediction separates the location of production from the location of organizational capture.

7.3 Scope and extensions

Several mechanisms are intentionally absent. Firm heterogeneity would turn the two organization thresholds into productivity cutoffs, as in trade intermediation models, but is not needed for the regional thresholds derived here. New Economic Geography feedback could make local service depth and market access endogenous to entry, possibly generating multiple equilibria. Dynamic endowment accumulation could move a region across industries over time. Platform strategic pricing could make the consumer and producer wedges jointly endogenous. These are natural extensions, but adding them to the present paper would obscure its distinct result: platform access can be diffuse while production and organizational capture remain structurally conditional.

8 Conclusion

Platform integration changes both sides of a local economy. It improves what households can buy, and it expands what local producers might sell. Those opportunities do not automatically become local development. Endowments determine latent comparative advantage; infrastructure determines whether an industry consistent with that advantage can cover its actual transaction and fixed costs and enter; and organizational choice determines whether complementary-service payments remain local.

This chain produces a platform-dependent intermediate state in which local production exists but organizational capture remains external. It also produces separate thresholds for nominal and consumption-equivalent local income. Consumer price gains make the real-income threshold weakly lower, while discrete entry or organization switches explain why strict ordering is conditional rather than universal. A facilitating state can help a potentially viable industry cross these thresholds by relaxing shared infrastructure financing constraints. A narrow external-channel restriction can also induce localization, but only by degrading an outside option and therefore cannot be treated as equivalent to cost-reducing facilitation.

The central conclusion is consequently not that platforms either help or hurt local development. It is that access and capture are different equilibrium objects. A common platform can equalize consumer access while leaving the production and organizational foundations of local income highly unequal.

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A Proofs

A.1 Consumer access and channel retention

Lemma 1 (Consumer access)**.** *Under equations* (2) *and* (3)*,*

$$s'(z) = (\eta - 1)s(z)[1 - s(z)] > 0 \quad \textit{and} \quad \frac{d \ln P_M(z)}{dz} = -s(z) < 0.$$

If $\ell_A > \ell_P$*, then* $\omega_C'(z) < 0$ *and* $\Lambda(z) > 0$*.*

Proof. Write:

$$x(z) = a_P p_P(z)^{1-\eta}, \qquad a = (1 - a_P)p_A^{1-\eta}.$$

Then $s = x/(a + x)$. Because $d \ln p_P/dz = -1$,

$$\frac{x'(z)}{x(z)} = -(1 - \eta) = \eta - 1.$$

Therefore:

$$s' = \frac{x'a}{(a + x)^2} = (\eta - 1)\frac{x}{a + x}\frac{a}{a + x} = (\eta - 1)s(1 - s) > 0.$$

For the price index:

$$\ln P_M = \frac{1}{1 - \eta} \ln(a + x).$$

Hence:

$$\frac{d \ln P_M}{dz} = \frac{1}{1 - \eta}\frac{x'}{a + x} = \frac{\eta - 1}{1 - \eta}s = -s.$$

Finally:

$$\omega_C'(z) = (\ell_P - \ell_A)s'(z).$$

It is negative when $\ell_A > \ell_P$. Since $D > 0$, the definition in equation (10) then implies $\Lambda > 0$. □

At the limits $s \to 0$ and $s \to 1$, $s' \to 0$. The platform continues to reduce P_M, but the expenditure-reallocation margin vanishes at either channel boundary. This is why capture drag is largest at an intermediate platform share rather than at complete platform penetration.

A.2 Endowment-conditioned production costs

The background technology in equation (12) implies:

$$w_i = (1 - \theta_0)\frac{Y_{i0}}{L_i}, \qquad r_i = \theta_0 \frac{Y_{i0}}{K_i}.$$

With $Y_{i0} = A_0 K_i^{\theta_0} L_i^{1-\theta_0}$, division by the relevant factor endowment yields equation (13). Substituting those factor prices into the candidate industry's Cobb–Douglas unit expenditure function gives:

$$\begin{aligned}
c_{ij}^P &= \frac{1}{A_j}\left(\frac{\theta_0 A_0 e_i^{\theta_0-1}}{\theta_j}\right)^{\theta_j}\left(\frac{(1-\theta_0)A_0 e_i^{\theta_0}}{1-\theta_j}\right)^{1-\theta_j} \\
&= \frac{A_0}{A_j}\left(\frac{\theta_0}{\theta_j}\right)^{\theta_j}\left(\frac{1-\theta_0}{1-\theta_j}\right)^{1-\theta_j} e_i^{\theta_0-\theta_j} \\
&\equiv \xi_j e_i^{\theta_0-\theta_j}.
\end{aligned}$$

Thus:

$$\frac{\partial \ln c_{ij}^P}{\partial \ln e_i} = \theta_0 - \theta_j.$$

For $\theta_j > \theta_0$, capital abundance lowers the candidate industry's production cost relative to the background opportunity cost. LCA still requires the external-reference double ratio in equation (16); the derivative is only the local structural component of that comparison.

A.3 Proof of Proposition 1

Proof. For a given organization r, equation (24) becomes:

$$\pi_r(z, G) = k\mathcal{M}_0 e^{(\sigma-1)(\psi z+\gamma G)}(c + d_r)^{1-\sigma} - F_r.$$

It is strictly increasing in G whenever $\Omega = (\sigma - 1)\gamma > 0$. The unique raw zero-profit threshold is equation (27). Direct differentiation gives:

$$\frac{\partial G_r^0}{\partial c} = \frac{\sigma - 1}{\Omega(c + d_r)} > 0.$$

The minimum of two strictly increasing functions is strictly increasing: for $c_2 > c_1$, both $G_P^0(c_2) > G_P^0(c_1)$ and $G_L^0(c_2) > G_L^0(c_1)$, so their minimum at c_2 exceeds their minimum at c_1. The truncation at zero preserves weak monotonicity. At an interior threshold, the relationship is strict.

Holding the external-region and background-industry ratios fixed, $\partial c/\partial \mathcal{C}_{ij}^P > 0$. The chain rule therefore gives:

$$\frac{\partial G_E}{\partial \mathcal{C}_{ij}^P} = \frac{\partial G_E}{\partial c}\frac{\partial c}{\partial \mathcal{C}_{ij}^P} > 0$$

at an interior threshold. Stronger LCA therefore starts from a lower production-cost ratio and requires weakly less G to cover all fixed and variable economic costs. Once the producer chooses an active mode, entry, production, and sales constitute the model's actual-production outcome, corresponding to the domestic-market realization of LCA discussed in the text. □

B Complete Entry and Organization Classification

Let:

$$b_P = k(c + d_P)^{1-\sigma}, \qquad b_L = k(c + d_L)^{1-\sigma},$$

so $b_L > b_P > 0$. It is convenient to express the state problem in terms of effective market scale:

$$X(z, G) = \mathcal{M}_0 e^{(\sigma-1)(\psi z + \gamma G)}.$$

Profits are:

$$\pi_P = Xb_P - f, \qquad \pi_L = Xb_L - f - F.$$

The corresponding market-scale thresholds are:

$$X_P = \frac{f}{b_P}, \qquad X_{L0} = \frac{f + F}{b_L}, \qquad X_X = \frac{F}{b_L - b_P}.$$

Lemma 2 (Three-region condition)**.** *Define:*

$$\bar{F} = f\frac{b_L - b_P}{b_P}.$$

If $F > \bar{F}$, then:

$$X_P < X_{L0} < X_X,$$

equivalently $G_P^0 < G_L^0 < G_X$, and the equilibrium states are:

$$\begin{cases} 0, & X < X_P, \\ P, & X_P \le X < X_X, \\ L, & X \ge X_X. \end{cases}$$

If $F = \bar{F}$, all three thresholds coincide. If $F < \bar{F}$, the platform-dependent interval is absent and the producer enters through L.

Proof. First:

$$\begin{aligned} X_P < X_{L0} &\iff \frac{f}{b_P} < \frac{f + F}{b_L} \\ &\iff f(b_L - b_P) < Fb_P \\ &\iff F > \bar{F}. \end{aligned}$$

Second:

$$\begin{aligned} X_{L0} < X_X &\iff \frac{f+F}{b_L} < \frac{F}{b_L - b_P} \\ &\iff f(b_L - b_P) < F b_P \\ &\iff F > \bar{F}. \end{aligned}$$

Thus the same condition orders all three thresholds. Below X_P, both profits are negative. At X_P, $\pi_P = 0$, while:

$$\pi_L - \pi_P = X_P(b_L - b_P) - F = \bar{F} - F < 0.$$

Because:

$$\pi_L - \pi_P = X(b_L - b_P) - F$$

is strictly increasing in X, it crosses zero exactly once at X_X. This proves the three states. Equality collapses all thresholds to one point. If $F < \bar{F}$, the local mode is already more profitable at the first profitable entry scale, so P is never selected. □

C Proof of Proposition 2

Proof. Define:

$$\Delta(c) = (c + d_L)^{1-\sigma} - (c + d_P)^{1-\sigma} > 0.$$

When $F > \bar{F}$, the raw switch threshold is:

$$G_X = \frac{\ln F - \ln[k\mathcal{M}_0 e^{(\sigma-1)\psi z}] - \ln \Delta(c)}{\Omega}.$$

Its cost derivative is:

$$\frac{\partial G_X}{\partial c} = -\frac{\Delta'(c)}{\Omega \Delta(c)}.$$

Because:

$$\Delta'(c) = (1-\sigma)\left[(c + d_L)^{-\sigma} - (c + d_P)^{-\sigma}\right] < 0,$$

we have $\partial G_X / \partial c > 0$.

When $F \leq \bar{F}$, Lemma 2 shows that the local mode enters directly, so the economically relevant embeddedness threshold is $\max\{0, G_L^0\}$, not G_X. Its derivative is:

$$\frac{\partial G_L^0}{\partial c} = \frac{\sigma - 1}{\Omega(c + d_L)} > 0.$$

Thus equation (34) is weakly increasing in c in either parameter region; truncation at zero preserves weak monotonicity. The LCA result follows from $\partial c / \partial \mathcal{C}_{ij}^P > 0$.

In the three-region case, for $G > G_X$:

$$\pi_L - \pi_P = k\mathcal{M}_0 e^{(\sigma-1)(\psi z+\gamma G)}\Delta(c) - F > 0.$$

Hence L is selected whenever it also weakly dominates no entry. In the direct-local case, it is selected once $G \geq G_L^0$. The classification in Lemma 2 therefore covers both possible entry orders.

Since $d_L < d_P$:

$$(c + d_L)^{1-\sigma} > (c + d_P)^{1-\sigma},$$

and therefore $\mathcal{R}_L > \mathcal{R}_P$ at a common (z, G). The payment-flow definitions imply:

$$\rho_L = 1 > 1 - \frac{d_P}{\mu(c + d_P)} = \rho_P.$$

Both inequalities are strict, so:

$$B_j^L = \rho_L \mathcal{R}_L > \rho_P \mathcal{R}_P = B_j^P.$$

The external payment changes from $d_P q_P > 0$ to zero in the polar baseline, while the unit organizational service $d_L q_L$ and the shared fixed organizational service F are locally supplied. The organizational switch therefore changes local service income, external payments, and total first-round local income. □

D Proof of Proposition 3

D.1 Income derivatives within a producer state

Fix an active producer organization r. The local producer-income term can be written:

$$B_j^r(z, G) = A_r \exp\{(\sigma - 1)(\psi z + \gamma G)\},$$

where:

$$A_r = \rho_r \mathcal{M}_0 \mu^{1-\sigma}(c + d_r)^{1-\sigma} > 0.$$

Therefore:

$$\frac{\partial \ln B_j^r}{\partial G} = \Omega > 0,$$

and:

$$\frac{\partial \zeta_r}{\partial G} = \Omega \zeta_r (1 - \zeta_r) > 0.$$

Define:

$$a_0 = (\sigma - 1)\psi > 0.$$

The nominal integration effect within organization r is:

$$h_Y(G) = \zeta_r(G)a_0 - \Lambda(z).$$

Its derivative is:

$$h'_Y(G) = \Omega\zeta_r(1-\zeta_r)a_0 > 0. \tag{59}$$

The price term $\alpha_M s(z)$ is independent of G, so:

$$h'_R(G) = h'_Y(G) > 0$$

within an active state. In the no-entry state, $\zeta = 0$; both effects are constant in G until entry.

D.2 Jumps at state changes

Immediately before entry, $\zeta = 0$, so $h_Y = -\Lambda$. Immediately after entry, the viable mode generates positive sales and local producer income, so $\zeta > 0$. Since $a_0 > 0$, the integration effect jumps strictly upward.

At a P-to-L switch, Proposition 2 gives $B_j^L > B_j^P$. Hence $\zeta_L > \zeta_P$, while a_0 and $\Lambda(z)$ are unchanged at the switch. Both nominal and real effects jump strictly upward.

D.3 Existence, uniqueness, and ordering

Completion of the proof. The preceding derivatives and jump results imply that the integration effects are globally nondecreasing in G, strictly increasing in every active organization, and have no downward state jumps. Under the baseline $\ell_A > \ell_P$ and an interior platform share, the no-entry effect is strictly negative because $\Lambda > 0$.

As $G \to \infty$, the local mode is selected and $\zeta_L \to 1$. Hence:

$$\lim_{G\to\infty} h_Y(G) = a_0 - \Lambda(z), \qquad \lim_{G\to\infty} h_R(G) = a_0 - \Lambda(z) + \alpha_M s(z).$$

Unlike a specification with an unbounded zG interaction, the parsimonious baseline does not make endpoint crossing automatic. A sufficient condition for a nominal threshold is $a_0 > \Lambda(z)$; a sufficient condition for a real threshold is $a_0 + \alpha_M s(z) > \Lambda(z)$, unless the real effect is already nonnegative at the lower endpoint. More generally, Proposition 3 states the required finite endpoint-crossing condition explicitly. Monotonicity and endpoint crossing imply a unique minimum threshold, allowing the crossing to occur either continuously or at an upward jump.

Pointwise:

$$h_R(G) = h_Y(G) + \alpha_M s(z) > h_Y(G).$$

Thus every G that makes the nominal effect nonnegative also makes the real effect positive, establishing $G_R \leq G_Y$.

Suppose both roots lie in the same active continuous organization and are interior. At G_Y,

$h_Y(G_Y) = 0$, so:

$$h_R(G_Y) = \alpha_M s(z) > 0.$$

Since h_R is continuous and strictly increasing in that organization, its zero must occur strictly below G_Y. Hence $G_R < G_Y$. If both effects jump from negative to nonnegative at the same discrete state change, their minimum thresholds are identical even though $h_R > h_Y$ on either side.

It remains to establish the comparative static with respect to LCA. For the local mode:

$$B_j^L = \mathcal{M}\mu^{1-\sigma}(c + d_L)^{1-\sigma},$$

which is strictly decreasing in c. For the platform mode:

$$B_j^P = \mathcal{M}\mu^{1-\sigma}\left[(c + d_P)^{1-\sigma} - \frac{d_P}{\mu}(c + d_P)^{-\sigma}\right].$$

Differentiating the bracket gives:

$$\begin{aligned}(1-\sigma)(c + d_P)^{-\sigma} &+ \frac{\sigma d_P}{\mu}(c + d_P)^{-\sigma-1} \\ &= -(\sigma - 1)c(c + d_P)^{-\sigma-1} < 0,\end{aligned}$$

where $\mu = \sigma/(\sigma - 1)$. Thus lower c raises local producer income in either mode. It also weakly advances entry and the switch to the higher-income local mode by Propositions 1 and 2. At every fixed G, stronger LCA therefore weakly raises ζ_{r^*} and the two integration effects. Their minimum thresholds weakly fall. The relationship is strict when the relevant root remains interior to the same active organizational state. □

E Proof of Proposition 4

Proof. Define the infrastructure provider's first-order condition:

$$\Phi(G, \vartheta) = \mathcal{U}_I'(G) - \kappa_I(\vartheta)C'(G) = 0.$$

Its derivative with respect to G is:

$$\Phi_G = \mathcal{U}_I''(G) - \kappa_I(\vartheta)C''(G) < 0.$$

The objective is therefore strictly concave and has at most one interior solution. Under the maintained endpoint conditions, that solution exists and is unique. Since:

$$\Phi_\vartheta = -\kappa_I'(\vartheta)C'(G) > 0,$$

implicit differentiation gives:

$$G'(\vartheta) = -\frac{\Phi_\vartheta}{\Phi_G} > 0.$$

For any fixed structural threshold G_H, the monotonicity of $G(\vartheta)$ implies that the set $\{\vartheta : G(\vartheta) \geq G_H\}$ is an interval. When nonempty, it has a unique minimum ϑ_H, except for an economically irrelevant flat boundary if the infrastructure solution itself is constrained. Increasing ϑ reduces $G_H - G(\vartheta)$. Stronger LCA lowers G_E, G_L, G_R, G_Y by Propositions 1–3; applying the increasing inverse of $G(\vartheta)$ weakly lowers the associated participation thresholds. □

E.1 External-channel restriction calculation

A consumer wedge $\tau_C > 1$ changes the effective platform price to $\tau_C \bar{p}_P e^{-z}$. The CES price index is increasing in an input price, so the restriction raises P_M and lowers the platform expenditure share. On the producer side:

$$\frac{\partial G_P^0}{\partial d_P} = \frac{\sigma - 1}{\Omega(c + d_P)} > 0.$$

Thus the restriction makes the low-fixed-cost external entry mode harder for a marginal producer.

Let:

$$\Delta = (c + d_L)^{1-\sigma} - (c + d_P)^{1-\sigma}.$$

Because:

$$\frac{\partial \Delta}{\partial d_P} = (\sigma - 1)(c + d_P)^{-\sigma} > 0,$$

the switch threshold satisfies:

$$\frac{\partial G_X}{\partial d_P} = -\frac{1}{\Omega \Delta} \frac{\partial \Delta}{\partial d_P} < 0.$$

The restriction can therefore induce a switch toward L, but it does so by worsening P, while simultaneously harming consumer access and making platform-based entry more difficult. Facilitation raises G, which increases market access under both modes and does not enter the consumer price equation. The two interventions are not equivalent.

F Boundary Cases and Counterexamples

F.1 No consumer-side capture gap

If $\ell_A = \ell_P$, then:

$$\omega_C(z) = \ell_A, \qquad \Lambda(z) = 0.$$

The nominal effect becomes:

$$\frac{d \ln Y}{dz} = \zeta_r(\sigma - 1)\psi \geq 0$$

in every active producer state. A negative nominal effect can then arise only in an extension with another first-round loss. This case confirms that the baseline negative region is generated by an explicit payment-incidence gap rather than platform terminology.

If $\ell_P > \ell_A$, then $\Lambda < 0$. Platform substitution raises consumer-side retention as well as reducing the price index. Both nominal and real effects shift upward; the low-infrastructure loss region may vanish.

F.2 No producer-side spatial payment gap

Setting $\lambda = 1$ in equation (37) gives $\rho_P = 1 = \rho_L$. The organization switch can still raise sales because $d_L < d_P$, but it no longer changes the spatial retention rate. This case separates the cost advantage of local organization from its payment-location advantage.

F.3 Positive infrastructure–platform complementarity extension

The baseline producer market access is:

$$\mathcal{M}(z, G) = \mathcal{M}_0 e^{(\sigma-1)(\psi z + \gamma G)}.$$

An optional extension adds $\chi^{\text{ext}} zG$ inside the final parentheses. The producer-access elasticity then becomes $(\sigma - 1)(\psi + \chi^{\text{ext}} G)$, so infrastructure raises both the level of candidate-industry income and its marginal response to platform integration. The baseline sets $\chi^{\text{ext}} = 0$ because this extra interaction is not needed for any core proposition; the numerical mechanism file shows how a positive value strengthens the producer-side effect.

F.4 Collapse of the platform-dependent region

Lemma 2 gives the exact counterexample to a universal three-stage development path. When $F \leq \bar{F}$, the local mode is sufficiently inexpensive in fixed-cost terms that the candidate industry enters directly through L. Platform dependence is therefore neither necessary nor universal.

F.5 Weak rather than strict income-threshold ordering

Suppose an entry or organizational switch occurs at $\widehat{G}$, with:

$$h_R(\widehat{G}^-) < 0, \qquad h_Y(\widehat{G}^-) < 0,$$

and:

$$h_R(\widehat{G}^+) \geq 0, \qquad h_Y(\widehat{G}^+) \geq 0.$$

Then:

$$G_R = G_Y = \widehat{G}.$$

The pointwise inequality $h_R > h_Y$ is not enough to create two different minimum thresholds when a common discrete jump crosses both zeroes. This is why Proposition 3 states strict ordering only under a continuous interior-root condition.

G Mapping to the Companion EL Income Block

The companion access–capture model can be written:

$$Y^{EL} = \frac{B^{EL} + \omega^{EL} G^{EL}}{1 - \beta^{EL} - \alpha^{EL}\omega^{EL}}.$$

The present paper does not insert this expression into the producer problem. It uses the following one-to-one mapping only after first-round payments have been determined:

Table 4: Mapping to the companion income-propagation block

Companion object	Present object	Accounting rule
$B^{EL} + \omega^{EL} G^{EL}$	$B_0 + B_j^{r^*}$	autonomous first-round local income only
α^{EL}	α_M	household market-good expenditure share
β^{EL}	β	locally supplied nontraded-good expenditure share
ω^{EL}	$\omega_C(z)$	local income content of induced market-good spending
consumer price index	$P_M(z)$	identical CES market-good price concept
nominal income	Y	equation (41)
consumption-equivalent income	R	equation (42)

The candidate-industry sales payment enters $B_j^{r^*}$ once. It does not also enter ω_C, which applies only to subsequent household market-good expenditure. The external platform organization payment enters ρ_P once. Consumer-channel external payments enter ℓ_P and ω_C, not ρ_P. This separation prevents production, organizational services, and induced household spending from being counted twice.

H Numerical Reproducibility and Mechanism Closures

The numerical implementation consists of:

- `code/numerical/model.py`, which implements the equations and equilibrium choices;
- `code/numerical/generate_figures.py`, which generates all figures and source-data CSV files;
- the validation script, which checks analytical derivatives, finite differences, threshold uniqueness, the three-region condition, boundary cases, and state monotonicity.

For $z = 0.5, c = 0.9$, the code obtains:

$$G_P^0 = 0.210619, \quad G_L^0 = 0.718802, \quad G_X = 1.090996,$$

$$G_R = G_E = 0.210619, \quad G_Y = 0.555415, \quad G_L = 1.090996.$$

The equality $G_R = G_E$ is event-exact: actual entry generates a discrete upward jump in the real effect. The threshold routine evaluates entry and organization events directly rather than returning the next point on a numerical grid. Finite-difference derivatives have the signs established in the proofs. The program also verifies that lowering F below $\bar{F}$ removes the platform-dependent interval and that equal consumer-channel local contents set Λ exactly to zero.

The file `figures/source_data/appendix_mechanism_closures.csv` reports the full G-grid for:

1. the baseline;
2. $\ell_A = \ell_P$;
3. $\ell_P > \ell_A$;
4. complete platform-service localization;
5. a positive access–infrastructure complementarity extension;
6. partial platform-service localization.

H.1 Equal-localization intervention comparison

The comparison holds $z = 0.5$ and $c = 0.9$ fixed. The baseline provides $G = 0.80$ and selects the external platform organization. The facilitation exercise raises shared infrastructure to $G = 1.15$. The external-channel-restriction exercise instead leaves $G = 0.80$, raises the external producer organization cost from $d_P = 0.40$ to 0.50, and applies a ten-percent consumer platform-price wedge. Both counterfactuals select L, so the comparison holds the binary localization outcome fixed rather than comparing different organizational regimes.

Table 5: Illustrative facilitation and external-channel restriction

Outcome	Baseline	Facilitation	Restriction
Infrastructure G	0.80	1.15	0.80
Equilibrium organization	P	L	L
Platform expenditure share	0.7655	0.7655	0.6696
Consumer price index	0.8287	0.8287	0.8875
Entry threshold G_E	0.2106	0.2106	0.7188
First-round local producer income	1.2485	3.9606	3.5658
Nominal local income	3.6516	8.0562	7.6657
Consumption-equivalent income	3.8998	8.6037	7.9926

Facilitation and the external-channel restriction both induce local organization in this illustrative comparison, but they do so through different margins. Facilitation leaves consumer access unchanged, expands producer access, and raises local producer income. The restriction lowers the consumer platform share, raises the price index, and more than triples the entry threshold relative to the unrestricted baseline. These results are source-data outputs rather than calibrated welfare estimates. The exact values are stored in `figures/source_data/appendix_channel_restriction.csv`.

All reported exercises use normalized, illustrative values. No parameter is estimated, and the exercises do not constitute a calibration.